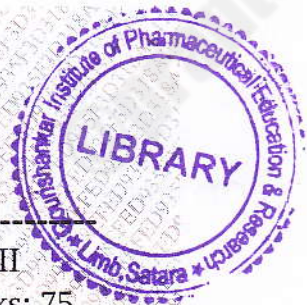


Dr. Babasaheb Ambedkar Technological University, Lonere
End Semester Examination (Summer), May 2017-18



Class: B. Pharm.

Semester: II

Subject with Subject Code: Human Anatomy and Physiology – II
[BP 201T]

Max. Marks: 75

Date: 22 / 05 / 2018

Duration: 3 Hrs.

Instructions:

1. All questions are compulsory.
2. Figure to the right indicates full marks.
3. Draw neat labelled diagram wherever necessary.

Q.1. Attempt the following multiple choice question (1×20)

20 Marks

- 1) Enzymes that digest fats have a general name
a) Lipase b) Trypsin c) Maltase d) Fatase
- 2) General name for enzymes which digest Proteins is
a) Pepsin b) Renin c) Maltase d) Protease
- 3) Mass movement in the colon after meal is due to the following except:
a) Gastrocolic reflex b) Duodenocolic reflex
c) Gastrin hormone d) Secretine hormone
- 4) Neuroglial cells support and protect
a) Muscle b) Nephron c) Glands d) Neuron
- 5) Molecules of neurotransmitter, released from the presynaptic neuron bind to receptors on the
a) Post synaptic membrane b) Cell body
c) Axonal membrane d) None of the above
- 6) During the propagation of a nerve impulse, the action potential results from the movement of
a) Na⁺ ions from extracellular fluid to intracellular fluid
b) K⁺ ions from extracellular fluid to intracellular fluid
c) Na⁺ ions from intracellular fluid to extracellular fluid
d) K⁺ ions from intracellular fluid to extracellular fluid
- 7) The vital centers for the control of visceral activities such as heart rate, breathing, blood pressure, swallowing, and vomiting are located in the:
a) Hypothalamus b) Pons c) Cerebrum d) Medulla oblongata
- 8) Bipolar neurons are commonly:
a) Motor neurons b) Found in the eye and nose
c) More abundant in adults than in children d) Found in ganglia

- 9) The gap between two communicating neurons is termed:
a) Node of Ranvier b) Synapse c) Effector d) Cell body
- 10) The substance that is released at axonal endings to propagate a nervous impulse is called:
a) An ion b) An action potential
c) The sodium-potassium pump d) A neurotransmitter
- 11) Which of the following hormone is involved in regulating water salt balance in blood
a) ADH b) ANH c) Growth hormone d) Cortisone
- 12) In male reproductive system, testicles are inside a sac called
a) Bladder b) Scrotum c) Rectum d) Glans
- 13) Primary female sex hormone produced by female reproductive system is
a) Gibberellins b) Oestrogen c) Adrenalin d) Thyroxine
- 14) Which part of the ovary act as an endocrine gland in human beings
a) Stroma b) Vitelline membrane c) Follicles d) Epithelium
- 15) Which malfunction of the respiratory system is the destruction of alveolar walls?
a) Bronchitis b) Pneumonia c) Emphysema d) Asthma
- 16) What is the function of insulin?
a) Antagonistic to glucagon b) Decreases glycogen storage in liver and muscle
c) Reduces hyperglycemia d) All of the above
- 17) Which of the following is the function of the lungs?
a) Metabolism
b) Serves as a reservoir of blood for the left ventricle.
c) It is a filter to protect the systemic vasculature
d) Facilitates the exchange of O_2 and CO_2 between air and blood.
- 18) Which statement is false?
a) The endocrine system is composed of ductless glands
b) Contents of the endocrine system are released into the bloodstream
c) The mammary gland is part of the endocrine system
d) Exocrine glands are not part of the endocrine system
- 19) What is the target of ACTH?
a) Mast cells b) Thyroid gland c) Mammary glands d) Adrenal cortex
- 20) The outer most covering of brain is
a) Dura Matter b) Pia matter c) Both a and b d) None of the above

20 Marks

Q.2 Solve any Two of the following:

- Explain in detail anatomy of neuron with structural and functional classification.
- Write in detail anatomy of stomach. Add a note on physiology of acid production.
- Explain in detail physiology of urine formation with a neat labeled diagram.

35 Marks

Q.3 Solve any Seven of the following.

- Explain in details various functions of liver.
- Draw a neat labeled diagram of spinal cord.
- Explain in details chemical digestion and absorption of nutrients in small intestine.
- Write a note on biosynthesis of thyroid hormone and add a note on BMR.
- Explain spermatogenesis or oogenesis.
- Write a note on anatomy of kidney.
- Explain in detail physiology of respiration.
- Write a note on protein biosynthesis.
- Explain in details different phases menstrual cycle with neat labeled diagram.





**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE -
RAIGAD -402 103**

Semester Winter Examination – Dec. - 2019

Branch: B. Pharmacy

Subject: Human Anatomy and Physiology II (BP201T)

Date:-03/12/2019

Sem.:- II

Marks: 75

Time:-3 Hr.

Instructions

1. All questions are compulsory
2. Figure to the right indicates full marks
3. Draw neat labelled diagram wherever necessary

Q.1. Attempt the following multiple choice question (1×20)

20 Marks

- 1) Enzymes that digest fats have a general name
a) Lipase b) Trypsin c) Maltase d) Fatase
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a) Muscle b) Nephron c) Glands d) Neuron
- 5) Molecules of neurotransmitter, released from the presynaptic neuron bind to receptors on the
a) Post synaptic membrane b) Cell body
c) Axonal membrane d) None of the above
- 6) During the propagation of a nerve impulse, the action potential results from the movement of
a) Na⁺ ions from extracellular fluid to intracellular fluid
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c) Na⁺ ions from intracellular fluid to extracellular fluid
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- 10) The substance that is released at axonal endings to propagate a nervous impulse is called:
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a) Bladder b) Scrotum c) Rectum d) Glans
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a) Mast cells b) Thyroid gland c) Mammary glands d) Adrenal cortex
- 20) The outer most covering of brain is
a) Dura Matter b) Pia matter c) Both a and b d) None of the above

Q.2 Solve any Two of the following (1×10)

20 Marks

- a) Explain in detail anatomy of neuron with structural and functional classification
- b) Write in detail anatomy of stomach. Add a note on physiology of acid production.
- c) Explain in detail physiology of urine formation with a neat labeled diagram.

Q.3 Solve any Seven of the following (7×5)

35 Marks

- a) Explain in details various functions of liver.
- b) Draw a neat labeled diagram of spinal cord.
- c) Explain in details chemical digestion and absorption of nutrients in small intestine.
- d) Write a note on biosynthesis of thyroid hormone and add a note on BMR
- e) Explain spermatogenesis or oogenesis
- f) Write a note on anatomy of kidney
- g) Explain in detail physiology of respiration
- h) Write a note on protein biosynthesis.
- i) Explain in details different phases menstrual cycle with neat labelled diagram.

Paper End



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: II

Subject Name: HUMAN ANATOMY AND PHYSIOLOGY-II

Subject Code: BP201T

Max Marks: 75

Date: 25/08/2022

Duration: 3.45 Hr.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = $20 \times 1 = 20$ (All the questions are compulsory)

- i) The alpha cells of Islets of Langerhans secrete
 - a. Glycogen
 - b. Insulin
 - c. Glucose
 - d. Glucagon
- ii) Bolus will be converted into chyme when it
 - a. Mix with saliva
 - b. Mix with gastric juice
 - c. Mix with amylase
 - c. Mix with intestinal juice
- iii) What is the major function of large intestine
 - a. Absorption of food
 - b. Absorption of water
 - c. Microbial activity
 - d. Propulsive activity
- iv) Sinusoid is the space present between
 - a. Two cells
 - b. Two lobes
 - c. Two plates
 - d. Cells within plate
- v) Trypsinogen is converted into trypsin in the presence of
 - a. Enterokinase
 - b. Gastric lipase
 - c. Enteric lipase
 - d. Trypsinase
- vi) "Cerebral aqueduct" is present between
 - a. Two lateral ventricles
 - b. Lateral ventricle and third ventricle



c. Third ventricle and fourth ventricle

d. Fourth and fifth ventricle

vii) When individual is lying on side, what will be the pressure of CSF?

a. 10 cm of water

b. 20 cm of water

c. 30 cm of water

d. 40 cm of water

viii) Unipolar neuron contains

a. One dendrite and one axon

b. One dendrite or one axon

c. Only dendrite

d. Only axon

ix) Which of the following is major neurotransmitter present in sympathetic nervous system?

a. Adrenaline

b. Nor-adrenaline

c. Dopamine

d. Acetylcholine

x) Fallopian tube is also known as

a. Uterine tube

b. Non-uterine tube

c. Oviducts

d. Both a and c

xi) Vital capacity is the sum of

a. Inspiratory reserve volume + Expiratory reserve volume

b. Inspiratory reserve volume + Expiratory reserve volume + Residual volume

c. Inspiratory reserve volume + Residual volume

d. Inspiratory reserve volume + Expiratory reserve volume + Tidal volume

xii) Transcription means

a. Conversion of DNA to Protein

b. Conversion of DNA to m-RNA

c. Conversion of m-RNA to Protein

d. Conversion of m-RNA to DNA

xiii) Which of the process is involve in biosynthesis of ATP?

a. Substrate level phosphorylation

b. Oxidative phosphorylation

c. Photophosphorylation

d. All of the above

xiv) Progesterone is secreted from

a. Endometrium

b. Myometrium

c. Corpus luteum

d. Corpus albicans

xv) Micturition process is controlled by

a. Voluntary nerves

b. In-voluntary nerves

c. Both a and b

d. None of the above

xvi) Which of the following is the secondary messenger?

a. c-AMP

b. G-Protein

c. d-ATP

d. t-RNA

xvii) Capillary hydrostatic pressure during filtration is built in the glomerulus as

a. size of Bowman's capsule is significantly large

b. an afferent arteriole is narrow compared to efferent

c. Bowman's capsule is cup-shaped

d. an efferent arteriole is narrow compared to afferent

xviii) Which of the following changes occur in diaphragm and intercostal muscles during expiration

a. Internal intercostal muscles relax and diaphragm contracts

b. External intercostal muscles relax and diaphragm contracts

c. Internal intercostal muscles contract and diaphragm relax

d. Internal intercostal muscles relax and diaphragm relax

xix) Hypersecretion of human growth hormone leads to

a. Dwarfism

b. Gigantism

c. Acromegaly

d. Both b and c

xx) The secretion of several anterior pituitary hormones are governed by other hormones from

a. Hypothalamus

b. Thyroid gland

c. Adrenal gland

d. Pancreatic lobes

Q. 2. Long Answers) = $2 \times 10 = 20$ (Answer 2 out of 3)

- i) Enlist all the organs of digestive system. Write a detail note on small intestine.
- ii) Draw neat labelled diagram of neuron and explain the neurophysiology.
- iii) Explain in detail physiology of urine formation.

Q. 3. Short Answers = $7 \times 5 = 35$ (Answer 7 out of 9)

- i) Describe the different roles of ATP.
- ii) Write a note on meninges.
- iii) Explain RAAS Pathway.
- iv) Write a note on posterior pituitary gland.
- v) Discuss the different phases of menstrual cycle.
- vi) Write a note on blood glucose regulation.
- vii) Explain different types of lung volumes and capacities.
- viii) Draw the neat labelled diagram of sperm and shortly explain spermatogenesis.
- ix) Discuss the translation process.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 03/05/2023

Course :	B. Pharmacy	Sem:	II
Subject Name :	Human Anatomy and Physiology-II	Subject Code :	BP201T
Max Marks :	75	Duration :	3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define the terms Synapse and Neuroglia.
- ii) Define Neurology and name the parts of CNS.
- iii) Define the terms Peristalsis and Achlorhydria.
- iv) Draw neat labeled diagram of Stomach.
- v) Define the terms Tidal Volume and Vital Capacity.
- vi) Define the terms glomerular filtration rate and Filtration fraction.
- vii) Define the terms hyperplasia and ectopic pregnancy.
- viii) Define the terms gynaecomastia and hypoglycaemia
- ix) What is cushing's syndrome.
- x) Enlist different phases of Menstruation Cycle.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Describe in detail about glomerular filtration and tubular reabsorption process.
- ii) Describe the structure and functions of Liver and explain various disease associated with it.
- iii) Discuss in detail female reproductive system.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Define action potential of neuron and explain in detail generation of nerve impulse.
- ii) Write a short note on hypothalamus.
- iii) Describe Functional Organization of Cerebral Cortex.
- iv) Draw the structure of testes and explain the spermatogenesis process.
- v) Discuss the mechanism of external respiration and factor affecting gaseous exchange.
- vi) Explain the process of absorption of food from small intestine.
- vii) Write a short note on Pituitary gland.
- viii) Describe the anatomy of thyroid gland and enlist name and functions of their hormones.
- ix) Discuss in detail the structure of neuron.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer-2023

Date:06/09/2023

Course : B. Pharmacy

Subject Name : Human Anatomy and Physiology-II

Max Marks : 75

Sem: II

Subject Code : BP201T

Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Draw well-labeled diagram of T.S. of small intestine.
- ii) What is artificial respiration? Mention the methods of artificial respiration.
- iii) What is Cushing's syndrome?
- iv) Mention the name of different parts of Nephron.
- v) Write any two examples of neurotransmitters.
- vi) Write note on cerebrospinal fluid
- vii) Discuss function of Digestive System
- viii) Define synapse and neuroglia
- ix) Enlist different phases of menstrual cycle.
- x) What is gynecomastia?

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Write Anatomy & Physiology of stomach. Explain Acid (HCl) production & secretion.
- ii) Explain in details female reproductive system.
- iii) Describe in details the physiology of Urine formation.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain RAAS pathway.
- ii) Write note on spermatogenesis.
- iii) Brief on Pituitary gland and hormone secreted by it.
- iv) Write Anatomy & Physiology of Lungs.
- v) Write brief note on spinal cord.
- vi) Enlist organ of Digestive system and its function.
- vii) Describe different role of ATP
- viii) Describe thyroid gland and hormone produced by it.
- ix) Discuss in details structure of Neuron.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: II

Subject Name: Human Anatomy & Physiology II

Subject Code: BP201T

Max Marks: 75

Date: 12-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10X2) =20
i)	Give functions of Hypothalamus	
ii)	Define Tidal volume & BMR	
iii)	Draw a neat labeled diagram of urinary system	
iv)	Enlist different functions of respiratory system	
v)	What do you mean by Pheochromocytoma and diabetes insipidus	
vi)	What are chromosomes	
vii)	Give composition & functions of bile juice	
viii)	What do you mean by neurogia & action potential	
ix)	Name hormones secreted by pancreas and give its functions	
x)	What are afferent and efferent nerve tract	
Q.2.	Long Answers (Answer 2 out of 3)	(10X2) =20
i)	Explain Oogenesis and Spermatogenesis	
ii)	Explain meaning of External, Internal respiration Give in detail physiology of respiration	
iii)	Explain in detail Physiology of Digestion of Fats, protein, Carbohydrates	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Write the functions CSF & show circulation of cerebrospinal fluid with help of diagram	
ii)	Explain different phases of menstrual cycle with neat labelled diagram	
iii)	Explain structure of Nephron with help of diagram	
iv)	Give functions of gastric juice & regulation of acid production in stomach	
v)	Write a short note on Cardiopulmonary resuscitation & Acute renal failure	
vi)	Enlist two functions of kidney & explain role of RAS Kidney	
vii)	Define Hormone and give its classification	
viii)	Write a note on structure and functions of spinal cord	
ix)	Name different hormone secreted by Pituitary gland & give its functions	
*** END OF THE PAPER ***		

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Course : B.Pharmacy

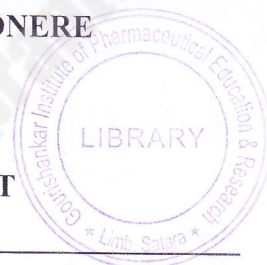
Sem: II

Subject Name: Human anatomy and physiology II

Subject Code: BP201T

Max Marks : 75

Duration :3Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

1. Solve the following. (Answer all the questions)

(10 x 2) = 20

- i. Draw neat labeled diagram structures representing digestive system.
- ii. Define the term tidal volume and vital capacity?
- iii. Define the term Hyperchlorhydria and Peristalsis?
- iv. Draw well labeled diagram of Neuron.
- v. Define Diabetes mellitus and give its types.
- vi. Mention different types of meninges present in brain.
- vii. Enlist different phase's menstrual cycle.
- viii. Enlist the parts of Urinary system.
- ix. Mention different hormones released by Pituitary gland.
- x. Write the composition of gastric juice.

2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i. Explain physiology of Urine formation with neat labeled diagram.
- ii. Discuss in detail male reproductive system and explain the process of spermatogenesis.
- iii. Explain the hormones released from pituitary gland with their role.

3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i. Explain the process chemical transmission of nerve impulses.
- ii. Explain the process of oogenesis.
- iii. Give the role of kidney in acid base balance.
- iv. Explain the process of absorption of food from small intestine.
- v. Explain the role of RAs in Kidney.
- vi. Explain what is BMR?
- vii. Explain Reflex action with neat labeled diagram.
- viii. Write the hormones released from pancreas and explain how they control blood glucose level.
- ix. What is Cardio Pulmonary Resuscitation? Why CPR is called as kiss of life?

-----END OF THE PAPER-----